Agenda:

1. Keys
   - → Candidate Keys
   - → Primary Keys
   - → Composite Keys
   - → Foreign Keys
2. Introduce SQL today

Candidate Keys

1. Students

name	email	phone	psp	b_id
Rahul	r@	706	89	1
Dheeraj	d@	034	90	2
Mohit	m@	678	89	1

Col Name	Super key	Candidate
name	✗	✗
email	✓	✓
phone Number	✓	✓
psp, batch	✗	✗
name, batch, psp	✗	✗
name, phone	✓	✗
email, phone, name	✓	✗
email, phone	✓	✗

either is sufficient to identify uniquely.

Candidate: Minimum (non redundant) cols which are required to uniquely identify a row

name, phone

~~email, phone~~

A

① students_attendance.

s_id	class_id	attendance
1	2	80
2	2	75
1	3	80

s_id	c_id
1,	1
1,	2
1,	3
2,	1

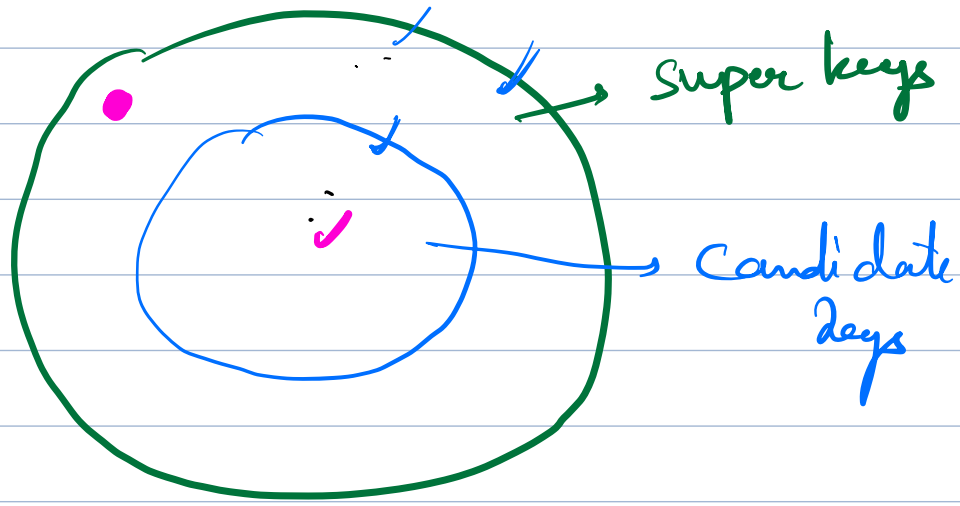
Q: Is s_id unique? ✗

Q: Is class_id unique? ✗

	S.K	C.K
s_id	✗	✗
class_id	✗	✗
attendance	✗	✗
s_id, class_id	✓	✓

Lovis	Maths
Lovis	Science
Lovis	Coding

~~s-id~~, ~~class-id~~

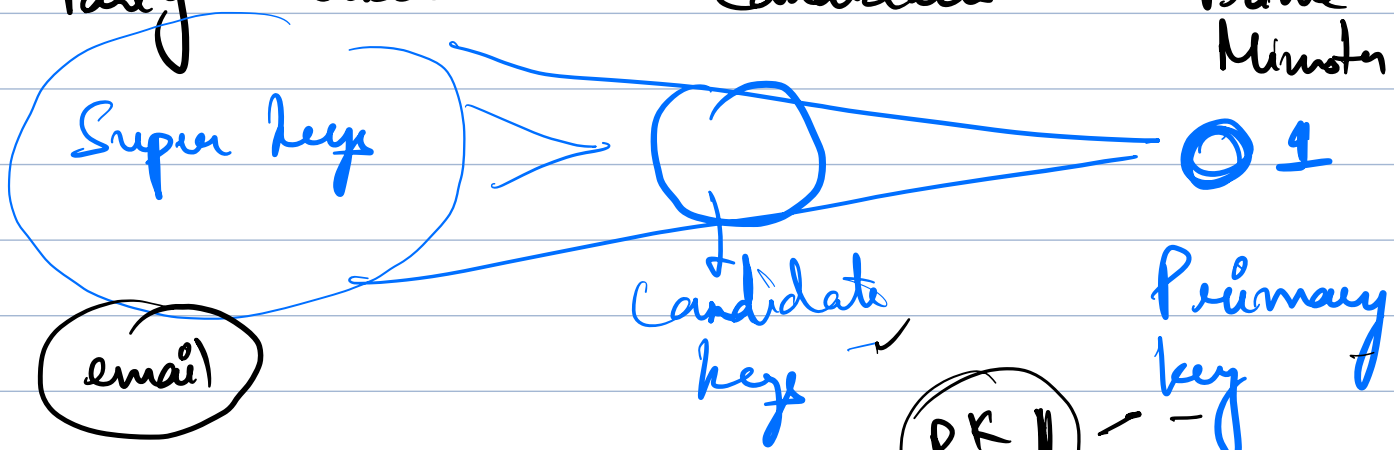


Emp Table.	SK	CK
Emp Id, depat	✓	✗
email	✓	✓
fNam, lLame	✗	✗
lnam, dep	✗	✗
emp Id, email	✓	✗

emp Id,	department
1	Math
2	Science
3	Math

Primary Keys

Party Workers → Candidates → Prime Minister (1)



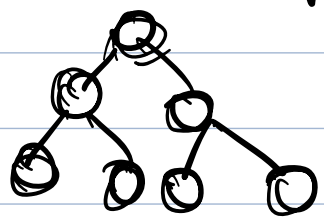
PK

emp_id	email	name
1	d@gm	d
2	a@gm	a
3	c@gm	c

(1) sort your data

(2) indexes on P.K

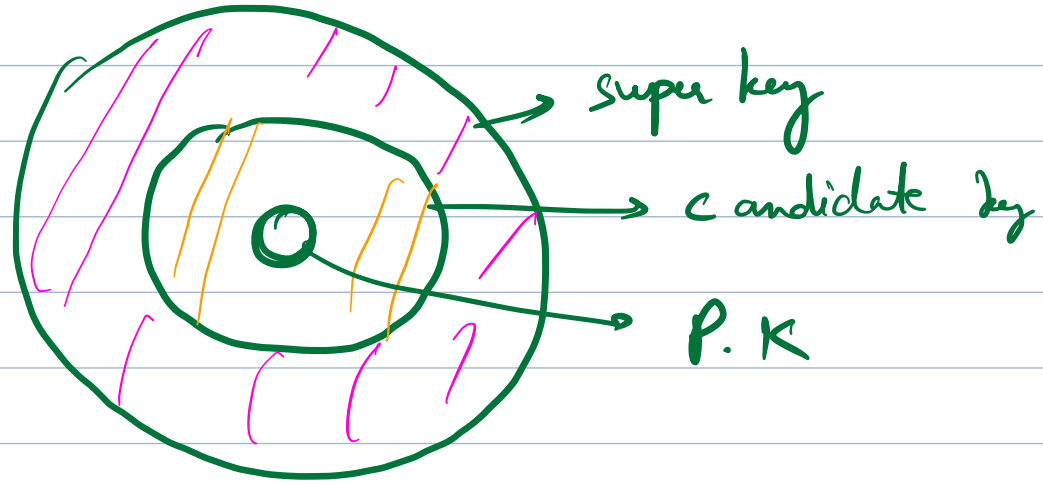
c@gmail



How to choose good PK.

(1) fast to sort
(2) smaller size
(3) Not get changed

id



Composite key

A key using more than 1 COLUMN to uniquely identify rows.

Intro to SQL

↳ Structured Query Language

Database ⟶ MySQL ⟶ SQL

⟵ RDBMS

emp_id	emp_name
1	~~dhruv~~ → human

Create ✓
Update ✓
Read ✓
Delete ✓

Create
↳ create a new DB
↳ create a new Table.

→ create Database nameOfDatabase;

→ create table tableName (name, email ---)

Foreign Keys.

relationship

Students

s_id	name	psp	b_id
1	Anmol	85	1
2	Sulaman	85	2
3	Dipesh.	85	1

Batch.

b_id	name
1	A
2	B
3	C

PK

Insert

foreign key

4, Sneha, 85, 4 ✗

delete Batch with b_id = 1
update Batch with b_id = 1 → 4

→ Cascade
→ Set Null
→ No action → default
→ Set Default

HW → MCQ's

① Students

s_id	name	email	phone	psp	b_id
1	Rahul	r@	706	89	1
2	Dheeraj	d@	034	90	2
3	Mohit	m@	678	89	1

Col N S 1 Cand

s_id, b_id SK
CK